



The Magnetic Map: How Sea Turtles Navigate Home

Grade 8 • North Carolina • Reading

RI.8.5

Analyze in detail how a specific paragraph in an informational text is structured, identifying the role each sentence plays in developing and refining a key concept. An 80-minute block-period packet with warm-up, guided practice, partner discussion, independent practice, small-group work, and a full answer key.

PRINT • TEACH • REVISIT



Plan the 80-minute block

Level and alignment

Grade 8 · North Carolina · Reading; RI.8.5 — Analyze in detail the structure of a specific paragraph in a text, including the role of particular sentences in developing and refining a key concept.

What is included

Cover; teacher guide; vocabulary and lesson preview; original passage; warm-up (2 items); guided practice with organizer (4 items); partner discussion (3 items); independent practice (6 items); small-group intervention and enrichment (4 items); exit ticket; complete answer key; access and sources page.

Suggested pacing (80 minutes)

Warm-up 8 min · passage + guided practice 25 min · partner discussion 12 min · independent practice 20 min · small-group work 10 min · exit ticket 5 min.

Preparation

Print student pages separately from the answer key. Review the passage and organizer before class. No outside novel or paid app required.

Student directions

Read the passage carefully. Use evidence from the text to support every answer. Quote or closely paraphrase the text and name the paragraph.



Before you read

Key vocabulary

Lesson focus

Every well-built paragraph has sentences that do different jobs: a topic sentence that introduces the key concept, supporting sentences that provide evidence or explanation, a sentence that refines or complicates the idea, and sometimes a transition sentence that connects to what follows. At grade 8, identify not just what a paragraph says, but what job each individual sentence is doing.

Model 1 — In a paragraph, the first sentence states, "Coral reefs face a growing threat from ocean acidification." What job is this sentence most likely doing?

It is most likely the topic sentence, introducing the paragraph's key concept (ocean acidification as a threat to coral reefs) that the rest of the paragraph will develop.

Model 2 — If a later sentence in that same paragraph says, "However, some coral species have shown surprising resilience in early trials," what job is that sentence doing relative to the topic sentence?

It is refining or complicating the topic sentence's claim, adding nuance by showing the threat is not uniform across all coral species, rather than simply repeating or supporting the original claim.



Warm-up · Before you read

Answer using what you already know. Five minutes.

1. What explicit fact does paragraph 1 state about the distance loggerhead turtles travel before returning to nest?

2. What is the topic sentence of paragraph 2, and what key concept does it introduce?



The Magnetic Map: How Sea Turtles Navigate Home (Part 1)

1. Every year, loggerhead sea turtles hatch on beaches along the southeastern United States, swim thousands of miles across the open Atlantic, and eventually return to nest on the very same stretch of coastline where they were born, sometimes after more than a decade at sea. For years, scientists had a name for this ability but not a full explanation: natal homing.
2. Researchers now believe the answer lies in a sense most humans do not have: magnetoreception, the ability to detect Earth's magnetic field. Every location on Earth has a slightly different magnetic signature, based on the angle and strength of the planet's magnetic field at that point. A sea turtle hatchling appears to imprint on the specific magnetic signature of its home beach within its first few weeks of life, the way some animals imprint visually on a parent.
3. That imprinted signature becomes something like an internal map. As the turtle travels thousands of miles over years, it can sense whether it is north, south, east, or west of its home signature, adjusting its swimming direction accordingly, the same way a hiker might use a compass without needing to see the destination directly.



The Magnetic Map: How Sea Turtles Navigate Home (Part 2)

4. A 2015 study tested this idea directly. Researchers exposed hatchling loggerheads to magnetic fields matching two different real locations along the Atlantic coast, then observed which direction the hatchlings tried to swim. The hatchlings consistently oriented themselves as if trying to move toward the coastline associated with their actual birth location, even though they had never experienced that magnetic field difference through any other sense.

5. The theory is not without complications. Magnetic signatures drift slightly over time due to natural shifts in Earth's magnetic field, meaning a turtle's internal map, formed once in infancy, could theoretically become slightly mismatched with its home beach's current signature decades later. Some researchers propose this drift may explain rare cases of turtles nesting on beaches near, but not exactly at, their birth location.

6. Despite these open questions, magnetoreception remains the leading explanation for one of the animal kingdom's most precise long-distance navigation feats, turning an ordinary beach into something closer to a magnetic address memorized before a turtle takes its first swim.



Guided practice · Evidence organizer

Work through this organizer together before independent practice.

It Says Quote or closely paraphrase the text.	It Means Explain the evidence in your own words.	And So State the inference or analysis it supports.

1. What job does the second sentence of paragraph 2 ("Every location on Earth has a slightly different magnetic signature...") perform relative to the topic sentence?

2. What job does the third sentence of paragraph 2 ("A sea turtle hatchling appears to imprint...") perform?

3. What is the topic sentence of paragraph 3, and how does it build directly on paragraph 2's final idea?

4. What job does the comparison to a hiker's compass, at the end of paragraph 3, perform in the paragraph's structure?

Partner discussion

Discuss each question with a partner before writing a joint response.

1. What is the topic sentence of paragraph 4, and what evidence does the rest of the paragraph provide to support it?

2. Which sentence in paragraph 4 states the study's specific result, and how does it connect back to the magnetoreception concept from paragraph 2?

3. What job does the first sentence of paragraph 5 ("The theory is not without complications.") perform in the paragraph's structure?

Independent practice

Work independently. Cite evidence from the text in every answer.

1. What specific complication does paragraph 5 introduce, and which sentence explains its possible real-world effect?

2. How does the first sentence of paragraph 6 ("Despite these open questions...") function as a transition, and what two ideas does it connect?

3. What job does the article's final phrase, comparing a beach to "a magnetic address memorized before a turtle takes its first swim," perform?



Independent practice

Work independently. Cite evidence from the text in every answer.

4. Choose paragraph 2 and explain, sentence by sentence, how each sentence builds toward a fuller understanding of magnetoreception.

5. Why might the article place the 2015 study's evidence (paragraph 4) after the explanation of magnetoreception (paragraphs 2–3) rather than before it?

6. Explain how paragraph 5's structure (introducing a complication, then explaining its possible effect) differs from paragraph 2's structure (introducing a concept, then explaining it directly). What is the purpose of each structure?



Small-group work

Use with students who need additional support.

1. If paragraph 5 (the complications) were removed entirely, how would that change the overall structure and credibility of the article?

2. What explicit fact does paragraph 1 state about how scientists previously understood natal homing?



Small-group work

Use with students ready for additional challenge.

1. How does the first sentence of paragraph 1 set up the concept that paragraph 2's topic sentence will answer?

2. What is the topic sentence of paragraph 6, and how does it function differently from the topic sentences of paragraphs 2 through 5?



Exit ticket

Answer independently to show what you learned today.

1. Explain how the structure of paragraphs 4 and 5 together mirrors the structure of scientific reasoning: presenting evidence, then testing its limits.



Exit ticket · self-assessment

Rate today's work:

4 — I can teach this to someone else.

3 — I can do this independently.

2 — I can do this with help.

1 — I need to revisit this with my teacher.



Answer key

1. They swim thousands of miles across the open Atlantic, sometimes returning after more than a decade at sea.
2. The topic sentence is "Researchers now believe the answer lies in a sense most humans do not have: magnetoreception," which introduces magnetoreception as the key concept the rest of the paragraph will explain.
3. It is a supporting sentence that provides background explanation needed to understand the topic sentence's claim, defining what a "magnetic signature" is before the paragraph can explain how turtles use it.
4. It is a supporting sentence that applies the general concept of magnetic signatures specifically to sea turtles, explaining how and when the imprinting process happens.
5. "That imprinted signature becomes something like an internal map" is the topic sentence; it builds directly on paragraph 2's idea of imprinting by explaining what the imprinted signature is then used for.

Accept equivalent, well-supported alternative answers throughout.



Answer key

6. It is a clarifying/supporting sentence that uses a familiar analogy to make the abstract concept of magnetic navigation easier to understand, refining the reader's understanding of the internal map concept.
7. "A 2015 study tested this idea directly" is the topic sentence; the rest of the paragraph provides the study's method (exposing hatchlings to different magnetic fields) and its result (hatchlings oriented toward their birth location) as supporting evidence.
8. "The hatchlings consistently oriented themselves as if trying to move toward the coastline associated with their actual birth location" states the result, directly confirming the magnetoreception and imprinting concepts introduced earlier in the article.
9. It is a refining/complicating sentence that signals a shift from supporting evidence to acknowledging limitations, preparing the reader for a more nuanced view of the theory.
10. The complication is that magnetic signatures drift over time; the sentence "Some researchers propose this drift may explain rare cases of turtles nesting on beaches near, but not exactly at, their birth location" explains its possible real-world effect.

Accept equivalent, well-supported alternative answers throughout.



Answer key

11. It connects the complications raised in paragraph 5 back to the overall strength of the magnetoreception theory, acknowledging uncertainty while reaffirming the theory's status as the leading explanation.
12. It is a concluding sentence that restates the article's key concept in a memorable, figurative way, giving the reader a final, vivid image that ties together imprinting, magnetic signatures, and navigation.
13. Sentence 1 introduces magnetoreception as the key concept; sentence 2 explains the background fact (magnetic signatures vary by location) needed to understand it; sentence 3 explains specifically how sea turtles use this ability (imprinting on their home signature), moving from general concept to specific mechanism across the three sentences.
14. Placing the explanation first gives the reader the background needed to understand why the study's results are significant; presenting the evidence first, without that context, would make the study's findings harder to interpret.
15. Paragraph 2 is structured to build understanding of a new concept step by step; paragraph 5 is structured to introduce doubt and then explain its implications, serving a different purpose — testing and qualifying the theory rather than simply explaining it.

Accept equivalent, well-supported alternative answers throughout.



Answer key

16. Removing paragraph 5 would make the article read as more one-sided and fully settled, losing the added credibility that comes from acknowledging open questions and showing the theory is still being tested and refined by researchers.

17. They had a name for the ability, "natal homing," but not a full explanation for how it worked.

18. Paragraph 1 ends by naming the unexplained phenomenon (natal homing) without a mechanism, and paragraph 2's topic sentence directly answers that gap by introducing magnetoreception as the likely explanation.

19. "Despite these open questions, magnetoreception remains the leading explanation..." is the topic sentence of paragraph 6; unlike the earlier topic sentences, which each introduce a new piece of the explanation or a complication, this one functions as a summary that reaffirms the overall theory after weighing its complications.

20. Paragraph 4 presents supporting evidence for the theory (the 2015 study's results), and paragraph 5 immediately follows by testing the theory's limits (the complication of magnetic drift), mirroring how real scientific reasoning builds support for an idea and then honestly examines where that idea might not fully hold.

Accept equivalent, well-supported alternative answers throughout.



Access, sources, and license

AIG

Ask for a second interpretation, a counterexample, or a student-written item at the same rigor, with an explanation of the choice.

504

Follow the individual plan: extended time, breaks, enlarged print, reduced copying, or an alternate response method as documented.

EC

Pre-teach key vocabulary before reading. Model the first guided-practice item aloud. Accept oral or dictated responses when appropriate.

ESL

Preview key vocabulary with examples. Allow brief home-language discussion before independent writing. Provide sentence starters for open-response items.

Official alignment source · accessed September 18, 2026

RI.8.5 — Analyze in detail the structure of a specific paragraph in a text, including the role of particular sentences in developing and refining a key concept.. The objective is original wording describing a focused part of the standard.

NC ELA Standard Course of Study, Grade 8: <https://www.thecorestandards.org/ELA-Literacy/RI/8/5/>

Original passage(s), questions, and answer explanations. No official released-test items or branded curriculum reproduced.

Print US Letter at actual size. Keep the answer key separate from student pages. Single-teacher classroom or LMS use; no resale or redistribution of complete files.